

4.2 Transformation of unit vectors

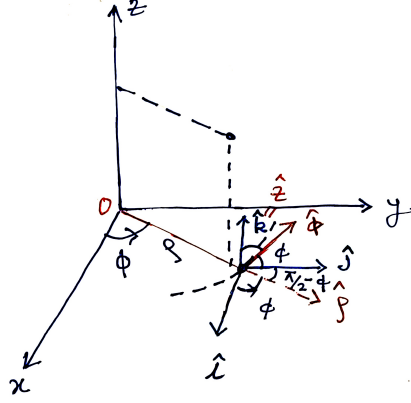


Figure 5: Cartesian and Cylindrical unit vectors.

The cylindrical unit vectors themselves may be expressed in the Cartesian basis. From the geometry of the angle ϕ we read off

$$\hat{\rho} = \cos \phi \hat{i} + \sin \phi \hat{j}, \quad (14)$$

$$\hat{\phi} = -\sin \phi \hat{i} + \cos \phi \hat{j}, \quad (15)$$

$$\hat{z} = \hat{k}. \quad (16)$$

The inverse relations are obtained by solving the linear system (or, equivalently, by forming the transpose of the orthogonal transformation matrix):

$$\hat{i} = \cos \phi \hat{\rho} - \sin \phi \hat{\phi}, \quad (17)$$

$$\hat{j} = \sin \phi \hat{\rho} + \cos \phi \hat{\phi}. \quad (18)$$

A Cartesian coordinate system is distinguished by the fact that its unit vectors are the same at every point of space. In polar or cylindrical coordinates the unit vectors $\hat{\rho}$ and $\hat{\phi}$ change direction from point to point; only $\hat{z}$ remains constant.

4.3 Position vector in cylindrical coordinates

$$\begin{aligned} \vec{r} &= x\hat{i} + y\hat{j} + z\hat{k} \\ &= \rho \cos \phi \hat{i} + \rho \sin \phi \hat{j} + z\hat{k} \\ &= \rho \cos \phi (\cos \phi \hat{\rho} - \sin \phi \hat{\phi}) + \rho \sin \phi (\sin \phi \hat{\rho} + \cos \phi \hat{\phi}) + z\hat{z} \\ &= \rho \cos^2 \phi \hat{\rho} - \rho \cos \phi \sin \phi \hat{\phi} + \rho \sin^2 \phi \hat{\rho} + \rho \sin \phi \cos \phi \hat{\phi} + z\hat{z} \\ &= \rho (\cos^2 \phi + \sin^2 \phi) \hat{\rho} + z\hat{z} \\ &= \rho \hat{\rho} + z\hat{z}. \end{aligned} \quad (19)$$

4.4 Transformation of vector components

Let a vector $\vec{A}$ possess Cartesian components (A_x, A_y, A_z) and cylindrical components (A_ρ, A_ϕ, A_z) . Substituting the expressions (14)–(16) into the expansion of $\vec{A}$ and collecting coefficients yields the linear relations

$$\begin{pmatrix} A_\rho \\ A_\phi \\ A_z \end{pmatrix} = \begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix}. \quad (20)$$

The inverse transformation is simply the transpose of the orthogonal matrix:

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} A_\rho \\ A_\phi \\ A_z \end{pmatrix}. \quad (21)$$

The determinant of the transformation matrix equals unity,

$$\det \begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix} = \cos^2 \phi + \sin^2 \phi = 1,$$

confirming that the transformation preserves orientation and volume.

5 Infinitesimal Elements in Cylindrical Coordinates

5.1 Infinitesimal line element

An infinitesimal displacement in cylindrical coordinates receives contributions from changes in each of the three coordinates. Geometry shows that a change $d\rho$ produces a radial displacement $d\rho \hat{\rho}$, a change $d\phi$ produces an arc-length displacement $\rho d\phi \hat{\phi}$, and a change dz produces a vertical displacement $dz \hat{z}$. Consequently

$$d\vec{r} = d\rho \hat{\rho} + \rho d\phi \hat{\phi} + dz \hat{z}. \quad (22)$$

The magnitude of the displacement is

$$|d\vec{r}| = \sqrt{(d\rho)^2 + (\rho d\phi)^2 + (dz)^2}.$$

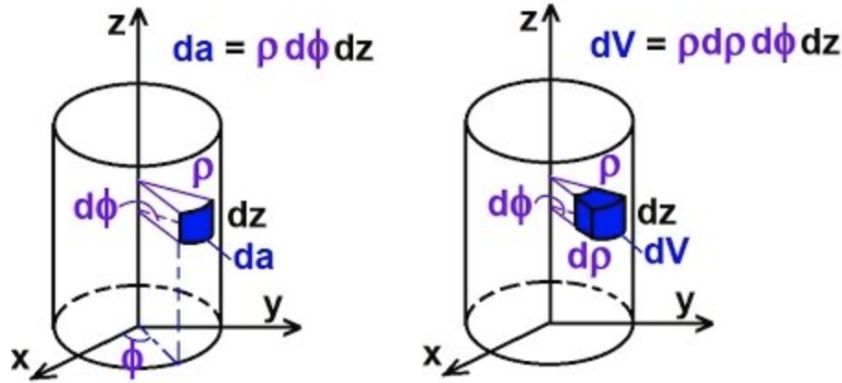


Figure 6: Infinitesimal displacements in the polar plane: radial increment $d\rho$ and azimuthal increment $\rho d\phi$.

5.2 Area elements

Three distinct oriented area elements appear in cylindrical coordinates.

- a) On the curved surface of a cylinder of radius ρ the area element is

$$d\vec{A}_\rho = \rho d\phi dz \hat{\rho}. \quad (23)$$

Its magnitude is the product of the arc length $\rho d\phi$ and the height dz .

b) On a horizontal disk (constant z) the area element is

$$d\vec{A}_z = \rho d\rho d\phi \hat{z}. \quad (24)$$

c) On a vertical half-plane of constant ϕ the area element is

$$d\vec{A}_\phi = d\rho dz \hat{\phi}. \quad (25)$$

5.3 Volume element

The volume of an infinitesimal cylindrical shell is the product of the three edge lengths:

$$dV = \rho d\rho d\phi dz. \quad (26)$$

The factor of ρ arises because the azimuthal edge has length $\rho d\phi$ rather than $d\phi$.

6 Example: Conversion of a Vector

Suppose a vector is given in Cartesian components by

$$\vec{A} = 3\hat{i} + 4\hat{j} + \hat{k}.$$

We first locate the point at which the vector is evaluated. The radial distance in the xy -plane is

$$\rho = \sqrt{3^2 + 4^2} = 5,$$

and the azimuthal angle satisfies

$$\phi = \tan^{-1}\left(\frac{4}{3}\right).$$

(The point lies in the first quadrant, so $0 < \phi < \pi/2$.) Applying the component transformation (20) we obtain

$$\begin{aligned} A_\rho &= A_x \cos \phi + A_y \sin \phi = 3 \cdot \frac{3}{5} + 4 \cdot \frac{4}{5} = \frac{9+16}{5} = 5, \\ A_\phi &= -A_x \sin \phi + A_y \cos \phi = -3 \cdot \frac{4}{5} + 4 \cdot \frac{3}{5} = \frac{-12+12}{5} = 0, \\ A_z &= 1. \end{aligned}$$

Thus in cylindrical coordinates the same vector reads

$$\vec{A} = 5\hat{\rho} + \hat{z}.$$

(The vanishing of the ϕ -component simply means that $\vec{A}$ points radially outward in the xy -plane.)

7 Summary

Quantity	Cartesian	Cylindrical
Position vector	$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$	$\vec{r} = \rho\hat{\rho} + z\hat{z}$
Line element	$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$	$d\vec{r} = d\rho\hat{\rho} + \rho d\phi\hat{\phi} + dz\hat{z}$
Volume element	$dV = dx dy dz$	$dV = \rho d\rho d\phi dz$
Unit-vector transformation	—	$\hat{\rho} = \cos \phi \hat{i} + \sin \phi \hat{j}$ $\hat{\phi} = -\sin \phi \hat{i} + \cos \phi \hat{j}$

The orthonormality relations, the transformation matrices for vector components, and the three distinct area elements complete the basic toolkit needed for calculations in cylindrical coordinates.

8 Spherical Polar Coordinates

Spherical polar coordinates are the natural choice whenever a physical problem possesses a degree of spherical symmetry. Any point P in three-dimensional space is uniquely labelled by the ordered triple (r, θ, ϕ) .

- r is the radial distance from the origin to the point P . Its range is

$$0 \leq r < \infty.$$

- θ is the polar angle (also called the colatitude or zenith angle). It is measured from the positive z -axis down to the position vector. Its range is

$$0 \leq \theta \leq \pi.$$

- ϕ is the azimuthal angle. It is measured in the xy -plane from the positive x -axis in the counterclockwise sense. Its range is

$$0 \leq \phi < 2\pi.$$

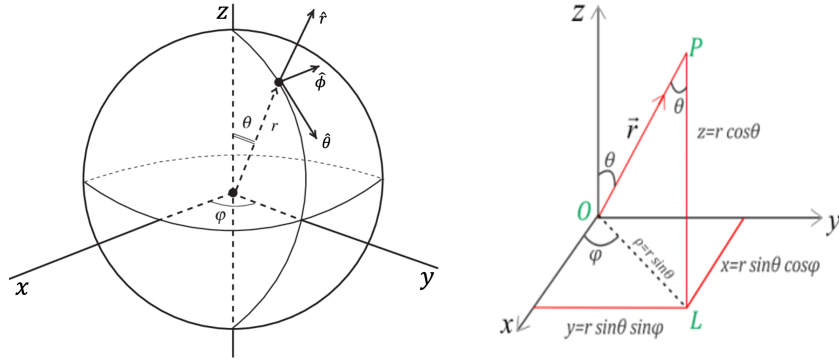


Figure 7: Spherical polar coordinates of a point P . The radial distance r runs from the origin to P , the polar angle θ is measured from the positive z -axis, and the azimuthal angle ϕ is measured from the positive x -axis in the xy -plane.

9 Unit Vectors in Spherical Coordinates

At every point we attach three mutually orthogonal unit vectors:

- $\hat{r}$ points in the direction of increasing r (radially outward),
- $\hat{\theta}$ points in the direction of increasing θ (tangentially downward),
- $\hat{\phi}$ points in the direction of increasing ϕ .

These unit vectors satisfy the orthonormality relations

$$\hat{r} \cdot \hat{r} = \hat{\theta} \cdot \hat{\theta} = \hat{\phi} \cdot \hat{\phi} = 1, \quad (27)$$

$$\hat{r} \cdot \hat{\theta} = \hat{\theta} \cdot \hat{\phi} = \hat{\phi} \cdot \hat{r} = 0, \quad (28)$$

together with the right-handed cross-product identities

$$\begin{aligned}\hat{r} \times \hat{\theta} &= \hat{\phi}, \\ \hat{\theta} \times \hat{\phi} &= \hat{r}, \\ \hat{\phi} \times \hat{r} &= \hat{\theta}.\end{aligned}\tag{29}$$

Because $\hat{r}$, $\hat{\theta}$ and $\hat{\phi}$ change direction from point to point, they are local bases; only at a fixed location do they behave as ordinary constant vectors.

10 Coordinate Transformations

10.1 From spherical to Cartesian coordinates

Given the spherical coordinates (r, θ, ϕ) of a point, the corresponding Cartesian coordinates follow at once from elementary trigonometry:

$$\begin{aligned}x &= r \sin \theta \cos \phi, \\ y &= r \sin \theta \sin \phi, \\ z &= r \cos \theta.\end{aligned}\tag{30}$$

The inverse transformation recovers the spherical coordinates from the Cartesian ones:

$$\begin{aligned}r &= \sqrt{x^2 + y^2 + z^2}, \\ \theta &= \cos^{-1}\left(\frac{z}{r}\right) = \tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right), \\ \phi &= \tan^{-1}\left(\frac{y}{x}\right).\end{aligned}\tag{31}$$

As always, the inverse tangent must be evaluated with attention to the quadrant determined by the signs of x and y .

10.2 Transformation of unit vectors

We first recall the cylindrical-to-Cartesian unit-vector relations

$$\begin{aligned}\hat{\rho} &= \cos \phi \hat{i} + \sin \phi \hat{j}, \\ \hat{\phi} &= -\sin \phi \hat{i} + \cos \phi \hat{j}, \\ \hat{z} &= \hat{k}.\end{aligned}\tag{32}$$

In the meridional plane the spherical unit vectors $\hat{r}$ and $\hat{\theta}$ are related to the cylindrical pair $(\hat{\rho}, \hat{z})$ by a simple rotation through the polar angle θ :

$$\begin{aligned}\hat{r} &= \sin \theta \hat{\rho} + \cos \theta \hat{z}, \\ \hat{\theta} &= \cos \theta \hat{\rho} - \sin \theta \hat{z}.\end{aligned}\tag{33}$$

(The azimuthal unit vector $\hat{\phi}$ is already common to both systems.) Substituting the expressions for $\hat{\rho}$ and $\hat{z}$ into (35) yields the complete transformation from the spherical basis to the Cartesian basis:

$$\begin{aligned}\hat{r} &= \sin \theta \cos \phi \hat{i} + \sin \theta \sin \phi \hat{j} + \cos \theta \hat{k}, \\ \hat{\theta} &= \cos \theta \cos \phi \hat{i} + \cos \theta \sin \phi \hat{j} - \sin \theta \hat{k}, \\ \hat{\phi} &= -\sin \phi \hat{i} + \cos \phi \hat{j}.\end{aligned}\tag{34}$$

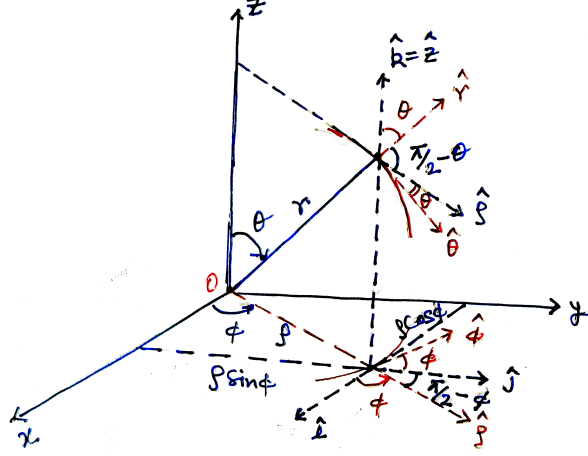


Figure 8: Unit vector transformation.

The inverse relations (Cartesian unit vectors expressed in the spherical basis) follow by solving the orthogonal system, or equivalently by taking the transpose of the transformation matrix:

$$\begin{aligned}\hat{i} &= \sin \theta \cos \phi \hat{r} + \cos \theta \cos \phi \hat{\theta} - \sin \phi \hat{\phi}, \\ \hat{j} &= \sin \theta \sin \phi \hat{r} + \cos \theta \sin \phi \hat{\theta} + \cos \phi \hat{\phi}, \\ \hat{k} &= \cos \theta \hat{r} - \sin \theta \hat{\theta}.\end{aligned}\tag{35}$$

10.3 Transformation of vector components

A general vector $\vec{A}$ expressed in the spherical basis takes the form

$$\vec{A} = A_r \hat{r} + A_\theta \hat{\theta} + A_\phi \hat{\phi}.\tag{36}$$

Its magnitude is therefore

$$|\vec{A}| = \sqrt{A_r^2 + A_\theta^2 + A_\phi^2}.\tag{37}$$

Let a vector $\vec{A}$ possess Cartesian components (A_x, A_y, A_z) and spherical components (A_r, A_θ, A_ϕ) . The component transformation is simply the matrix whose rows are the Cartesian components of the spherical unit vectors:

$$\begin{pmatrix} A_r \\ A_\theta \\ A_\phi \end{pmatrix} = \begin{pmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix}.\tag{38}$$

The inverse transformation is the transpose of the same orthogonal matrix:

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{pmatrix} \begin{pmatrix} A_r \\ A_\theta \\ A_\phi \end{pmatrix}.\tag{39}$$

Because the matrix is orthogonal its determinant equals ± 1 ; explicit evaluation shows that the determinant is $+1$, confirming that the transformation preserves orientation.